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Unit-4
Assignment-4

Section-A

Q.1) What are the data structures that are used for clustering?

Ans) The main data structures used for clustering include:

- Data Matrix: rows represent objects, columns represent attributes.
- Dissimilarity (distance) matrix: Stores pairwise distance between objects.
- Tree Structures - e.g., dendograms used in hierarchical clustering
- Grid-based structures - Spatial partitioning for grid-based clustering
- CF-tree (Clustering Feature tree) - used in BIRCH algorithm.

Q.2) Discuss various issues regarding classification.

Key issues in classification:

- Data quality issues (noise, missing values, outliers)
- Overfitting and Underfitting
- Imbalanced class distribution
- Selection of relevant features
- Scalability and efficiency
- Interpretability of the model

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Q.3) Write the formula for arithmetic distance used in K-Means algorithm

Ans) K-Means uses Euclidean distance to measure similarity:

$$d(x, c) = \sqrt{\sum_{i=1}^{n} (x_i - c_i)^2}$$

where
x = data point
c = cluster centroid

Q.4) What does BIRCH stand for?

Ans) BIRCH stands for:
Balanced Iterative Reducing and Clustering using Hierarchies.

Q.5) What is a dendrogram?

Ans) A dendrogram is a tree-like diagram that visualizes the process of hierarchical clustering. It shows how clusters are merged or split step by step and helps determine the cluster structure.

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Section - B

Q.1) Write a short note on DBSCAN algorithm?

Ans) DBSCAN stands for Density - Based Spatial Clustering of Applications with Noise.
It is a density - based clustering algorithm that groupes together data points that are closely packed and marks points in low - density regions as outliers / noise.

Key Concepts

- ε (Epsilon) : Neighbourhood radius around a point.
- MinPts : Minimum number of points required to form a dense region.

Point Categories

- Core Point : Has at least MinPts within ε radius.
- Border Point : Lies within ε of a core point but has fewer than MinPts.
- Noise Point : Neither core nor border point; considered an outlier.

How DBSCAN Works

1. Select an unvisited point.
2. If the point has ≥ MinPts within ε → mark as core point and create a cluster.
3. Expand cluster by recursively including all density - reachable points.
4. If point does not meet criteria → mark as noise

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5. Continue until all points processed.

Advantages
- Can find clusters of arbitrary shape
- Handles outliers effectively
- No need to specify number of clusters beforehand.

Disadvantages
- Struggles with varying density clusters
- Performance sensitive to parameter ε and Min Pts.

Applications
- Spatial data analysis
- Image segmentation
- Fraud detection
- Anomaly Detection.

Q.2) What are data points and their types?

Ans) Data Point
- A data point represents a single record or observation in a dataset.
- It consists of features / attributes and is used in clustering or classification tasks.

Example:
Customer record
{ Age = 30, Income = 50k, City = NY }

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Types of Data Points

1. Numerical Data Points
   - Have numerical values
   - Used in statistical analysis
   - Example : Height = 165 cm, Salary = 60000

2. Categorical Data Points
   - Represent categories / labels
   - Example : Gender = Male, Color = Red

3. Binary Data Points
   - Only two possible values (0/1, Yes/No)
   - Example : Passed = 1, Failed = 0

4. Ordinal Data Points
   - Ordered Categories
   - Example : Education level (Primary < Secondary < Graduate)

5. Time-Series Data Points
   - Values recorded over time
   - Example : Stock prices per day

6. Spatial Data Points
   - Represent location or coordinates
   - Example : GPS point (Lat, Long)

7. Text / Document Data
   - Unstructured textual records

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Role in Clustering
- Data points are grouped based on similarity
- Distance measures (eg., Euclidean) determine closeness.

Model Based Cluste

Q.3> What is clustering and its types?

Ans> Clustering is an unsupervised machine learning technique that groups similar data points into clusters where objects in the same cluster are more similar to each other than to those in other clusters.

Types of Clustering

Type	Description	Example
Partitioning Clustering	Divides data into k clusters	K-means, K-medoids
Hierarchical Clustering	Builds tree-like structure	Agglomerative and Divisive clustering
Density-Based Clustering	Groups points based on density.	DBSCAN, OPTICS
Grid-Based Clustering	Data space divided into grids	STING, CLIQUE

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Model-Based Clustering	Assumes data follows a statistical model — EM Algorithm, Gaussian Mixture Models.

Applications
- Market segmentation
- Image Processing
- Customer Behaviour Analysis
- Social Network Analysis
- Medical diagnosis

Section - C

1.) Explain K-Means algorithm with suitable example.

Ans.) K-Means is a partitioning-based clustering algorithm that divides data into K clusters, where each record belongs to the cluster with the nearest mean (centroid).

Steps of K-Means Algorithm

1. Choose K (number of clusters)
2. Initialize K centroids randomly
3. Assign each data point to the nearest centroid (using Euclidean distance)
4. Recalculate centroids by finding mean of assigned points
5. Reassign points based on new centroids
6. Repeat steps 4-5 until
   - Centroids do not change or
   - Maximum iteration reached

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Euclidean Distance Formula

$$d(x, c) = \sqrt{\sum_{i=1}^{n} (x_i - c_i)^2}$$

Example

Dataset: 6 points →

(2, 10), (2, 5), (8, 4), (5, 8), (7, 5), (6, 4)

Choose K = 2

Initial Centroids

$C_1 = (2, 10)$

$C_2 = (8, 4)$

Step 1: Assign points to nearest centroid

Point	Dist to C_1	Dist to C_2	Cluster
(2,10)	0	7.2	C1
(2,5)	5	6.3	C1
(8,4)	8.9	0	C2
(5,8)	4.2	4.5	C1
(7,5)	6.4	1	C2
(6,4)	7.2	2	C2

Clusters formed:

- Cluster 1: (2,10), (2,5), (5,8)
- Cluster 2: (8,4), (7,5), (6,4)

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Step 2: Recalculate Centroids

$$C_1 = \left(\frac{2+2+5}{3}, \frac{10+5+8}{3}\right) = (3, 7.7)$$

$$C_2 = \left(\frac{8+7+6}{3}, \frac{4+5+4}{3}\right) = (7, 4.3)$$

Repeat assignment until Convergence

Final result: 2 clusters with updated centroids

Advantages
- Simple and efficient
- Works well for large datasets

Limitations
- Must choose k in advance.
- Sensitive to outliers.
- Works well only for Spherical clusters

Q.2) Differentiate between methods of hierarchical clustering.

Ans) Hierarchical clustering builds a tree-like cluster structure (dendrogram).

Two approaches:

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Feature	Agglomerative (Bottom-up)	Divisive (Top-down)
Process	Starts with individual points and merges clusters.	Starts with one large cluster and splits it.
Beginning	n clusters → 1 cluster	1 cluster → n clusters
Method	Repeated merging based on similarity	Repeated splitting based on dissimilarity
Strategy	Greedy Approach	Computationally Complex
Popularity	Most commonly used	Rare in practical use
Time Complexity	$O(n^2)$	Higher than agglomerative
Dendrogram Direction	Leaves → Root	Root → Leaves

Linkage Methods (Agglomerative)

Method	Distance Definition
Single Linkage	Nearest Neighbour
Complete Linkage	Farthest Neighbour
Average Linkage	Average distance between clusters
Centroid Linkage	Distance between centroids

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Ward's Method — Minimizes variance within clusters.

Advantages
- No need to pre-set number of clusters
- Produces dendrogram for visualization.

Disadvantages
- Computationally expensive
- Difficult to handle large data

Q.3) What do you mean by outlier analysis? Briefly describe types of outliers.

Ans) Outlier analysis identifies data points that do not conform to normal behaviour.
Outliers are unusual, rare or inconsistent observations.

Need for Outlier Analysis
- Fraud / anomaly detection
- Medical data diagnosis
- Network intrusion detection
- Removing noise for ML models

Types of Outliers

Type	Description	Example
Global (Point) Outliers	Individual data point deviates significantly from rest	Salary 10M in a dataset of 30k - 80k

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Contextual Outliers	Unusual only in a Specific Context	Temperature 20°C normal in summer but outlier in winter
Collective Outliers	A group of related data points abnormal together	Sudden spike in system traffic → cyber attack

Outlier Detection Methods

1. Statistical Methods
   z-score, box-plot, IQR
2. Distance-based Methods
   k-NN distance, clustering-based
3. Density-based Methods
   LOF (Local Outlier Factor), DBSCAN
4. Machine Learning Methods
   SVM, Isolation Forest, Autoencoders

Applications

- Banking fraud detection
- Medical anomaly detection
- Network security
- Quality control in industry

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